HARDWARE INTEGRATION & CONFIGURATION

Thermal POS Printer (80mm) & Windows Setup Manual

Step-by-step engineering manual for installing 80mm and 58mm thermal receipt printers, ESC/POS hardware command sequencing, cash drawer kickout, and silent browser printing.

CLASSIFICATION	Hardware Deployment	RELEASE VERSION	v2.4.0 (Enterprise)
AUTHOR / TEAM	Nexus Hardware Integration Lab	EFFECTIVE DATE	October 2026
TARGET AUDIENCE	IT Support Technicians, System Integrators, POS Cashiers	STATUS	Official Production Ready

1. Hardware Specifications & Interfaces

Nexus ERP Web POS utilizes high-performance browser-direct printing compatible with all standard 80mm and 58mm ESC/POS thermal printers (including Xprinter, Rongta, Epson TM-T82, Posiflex, Bixolon, and generic OEM USB receipt printers).

Parameter	80mm Standard Specification	58mm Compact Specification
`Print Method`	Direct Thermal Line Printing	Direct Thermal Line Printing
`Paper Width`	79.5mm ± 0.5mm	57.5mm ± 0.5mm
`Printable Width`	72mm (576 dots/line)	48mm (384 dots/line)
`Resolution`	203 DPI (8 dots/mm)	203 DPI (8 dots/mm)
`Interface Types`	USB 2.0, Ethernet (LAN), Bluetooth	USB 2.0, Bluetooth
`Cash Drawer Port`	RJ11/RJ12 24VDC 1A	RJ11 12VDC

2. Windows Driver Installation & Port Binding

Follow these steps to establish clean Windows driver binding prior to configuring browser printing:

Step 1: Connect the thermal printer to a USB port and power on the device. Insert the 80mm thermal paper roll with the coated thermal side facing upward.

Step 2: Install the manufacturer's Windows driver (e.g., *POS-80 Driver Setup* or *Epson Advanced Printer Driver*).

Step 3: Open **Windows Control Panel** → **Devices and Printers**. Right-click the installed printer and select **Printer Properties**.

Step 4: Navigate to the **Ports** tab. Verify that the printer is bound to the active `USB001` or `USB002` virtual printer port.

Step 5: In the **General** tab, click **Print Test Page**. Verify that the printer outputs the Windows test banner cleanly.

3. Configuring Page Margins & Paper Sizes

By default, Windows drivers often assume Letter or A4 dimensions. You must configure the paper size to continuous thermal roll dimensions:

- In Printer Properties, open **Preferences** → **Advanced**.
- Under Paper Size, select **80mm x 297mm** or **80mm x Continuous**.
- Under Layout Margins, set **Left: 0mm, Right: 0mm, Top: 0mm, Bottom: 0mm**.

Cash Drawer Automatic Kickout Configuration

To make the cash drawer pop open automatically upon bill completion, navigate to: Printer Properties → Device Settings → Cash Drawer. Select 'Open Before Printing' or 'Open After Printing' for Drawer #1. This sends the standard ESC/POS pulse (ESC p 0 25 250) down the RJ11 cable.

4. High-Speed Kiosk Silent Printing Setup

To allow instant 1-second invoice printing without displaying the Windows print confirmation popup, configure your browser in **Kiosk Mode**:

1. Right-click your Chrome/Edge desktop shortcut and select **Properties**.
2. In the **Target** input field, append the flag: `--kiosk-printing` to the end of the line.
   Example: `"C:\Program Files\Google\Chrome\Application\chrome.exe" --kiosk-printing`
3. Set your 80mm POS printer as the **Default Windows Printer**.
4. In Nexus POS, pressing **F12 / Print** will now fire the invoice to the paper roll instantly with zero dialog delay.

5. Troubleshooting Common Hardware Faults

Symptom	Root Cause	Resolution Action
`Blank paper feeds without text`	Roll inserted upside down	Reverse roll so thermal coated side faces printhead.
`Cutter fails to sever paper`	Paper jam or worn blade	Open cutter cover, turn manual dial to release blade.
`Cash drawer does not kick`	RJ11 cable loose or wrong voltage	Check RJ11 cable; ensure 24V solenoid compatibility.
`Faint / blurry barcode printing`	Low heat density or dirty head	Clean head with isopropyl alcohol; increase darkness in driver.